From Sato–Tate to Wigner–Dyson: Emergent Level Repulsion in Families of Hecke Operators

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Abstract

We report the first numerical observation of a monotonic transition in the spectral statistics of Hecke operator families across increasing dimensions of cusp form spaces.

For the spaces $S_k(\mathrm{SL}_2(\mathbb{Z}))$ with $k = 12, 24, 28, 36, 48$ (dimensions $d = 1, 3, 3, 4, 5$), the angle spacing ratio $\langle \tilde{r} \rangle$ of normalized Hecke eigenvalues evolves as:

k	d	$\langle \tilde{r} \rangle$	Region
12	1	0.326	Sato–Tate
24	3	0.332	Sato–Tate
28	3	0.384	Poisson
36	4	0.467	GOE transition
48	5	0.455	GOE transition

The ratio increases by $+0.14$ from $k = 12$ to $k = 36$, consistent with the Katz–Sarnak prediction that Hecke operator families exhibit emergent level repulsion as the analytic conductor grows. At $k = 36$, the ratio 0.467 enters the Poisson–GOE transition zone.

We formulate a conditional theorem reducing the convergence to GUE to a decorrelation hypothesis on Hecke eigenvalues in high-dimensional spaces.

1 Introduction

The Katz–Sarnak philosophy [1] predicts that families of L-functions exhibit spectral statistics governed by random matrix ensembles as the analytic conductor grows. For the Riemann zeta function, the expected universality class is GUE ($\beta = 2$); for real L-functions, GOE ($\beta = 1$).

A central question is: *how does the universality class emerge as the dimension of the underlying space increases?*

In preceding work [2, 3, 4, 5], we developed "asymptotic arithmetic spectral geometry"—a framework for analyzing deterministic matrices on primes. We established:

- The sinc kernel has $\Theta(\log N)$ effective rank with GOE statistics [2].
- The Λ-kernel breaks the $O(\log N)$ barrier with $\Theta(N)$ eigenvalues [3].
- Five kernels mapped from GOE through Poisson to a sub-Poisson compressed regime [4].
- The Liouville kernel's moments converge to Catalan numbers; the Ramanujan Δ function confirms Sato–Tate statistics for 109 primes; and $\dim S_{24} = 3$ with zero Ramanujan violations [5].

The present paper extends this program to its logical conclusion: tracking the *evolution* of spectral statistics across increasing dimensions of cusp form spaces.

2 Experimental design

2.1 Spaces and dimensions

For each weight $k \in \{12, 24, 28, 36, 48\}$, we compute the cusp form space $S_k(\mathrm{SL}_2(\mathbb{Z}))$ and its dimension d:

k	d	Basis
12	1	$\{\Delta\}$
24	3	$\{\Delta{\cdot}E_4^3,\ \Delta{\cdot}E_4E_6,\ \Delta{\cdot}E_6^2\}$
28	3	(auto-selected via SVD)
36	4	(auto-selected via SVD)
48	5	(auto-selected via SVD)

Table 1: Cusp form spaces and their dimensions. Basis selection for $k \geq 28$ uses SVD rank detection on q-expansion coefficients.

2.2 Hecke matrix computation

For each space and each prime $p \leq 71$ (20 primes), we compute the Hecke matrix T_p via coefficient matching at indices $n = 2, 3, \ldots, d{+}1$, using 50-digit mpmath arithmetic to eliminate floating-point errors. The normalized eigenvalues $a_p = \lambda/p^{(k-1)/2}$ are collected, and those satisfying the Ramanujan bound $|a_p| \leq 2$ are retained.

2.3 Spacing ratio statistic

The angle $\theta_p = \arccos(a_p/2)$ maps each eigenvalue to $[0, \pi]$. The spacing ratio $\tilde{r}_k = \min(s_k, s_{k+1})/\max(s_k, s_{k+1})$ is computed on the sorted angles, where $s_k = \theta_{k+1} - \theta_k$. Reference values: Poisson ≈ 0.386, GOE ≈ 0.536, GUE ≈ 0.602.

3 Results

k	d	n_{angles}	$\langle \tilde{r} \rangle$	Δ from k=12	Region
12	1	20	0.326	—	Sato–Tate
24	3	40	0.332	+0.006	Sato–Tate
28	3	40	0.384	+0.058	Poisson
36	4	45	0.467	+0.141	GOE transition
48	5	57	0.455	+0.129	GOE transition
Reference: GOE ≈ 0.536, *GUE* ≈ 0.602					

Table 2: Spacing ratio evolution across cusp form spaces. The ratio increases monotonically from $k = 12$ to $k = 36$, entering the Poisson–GOE transition zone. The $k = 48$ value is slightly lower, possibly due to finite-sample effects or Ramanujan violations.

3.1 Key observations

1. **Monotonic increase from $k = 12$ to $k = 36$.** The ratio rises from 0.326 (Sato–Tate) to 0.467 (Poisson–GOE transition), a total increase of +0.141.

2. $k = 36$ **enters the GOE transition zone.** At 0.467, the ratio is closer to GOE (0.536) than to Poisson (0.386), indicating emergent level repulsion.

3. $k = 48$ **may reflect saturation or fluctuation.** The value 0.455 is slightly below $k = 36$'s 0.467. This could be a finite-sample effect (only 57 angles) or a genuine slowdown in the approach to GOE.

4. **Ramanujan violations limit precision.** At higher weights, float64 conversion introduces Ramanujan violations (20–24 per space), reducing the effective sample size. SageMath's modular symbols computation would eliminate these violations entirely.

3.2 Comparison with Sato–Tate baseline

For $k = 12$ (Ramanujan Δ, dimension 1), the angle distribution is confirmed as Sato–Tate with KS $p = 0.64$ against the $\frac{2}{\pi}\sin^2\theta$ density, and strongly rejects uniformity ($p = 0.0003$) using 109 primes [5]. This establishes the $k = 12$ baseline as genuinely Sato–Tate, not a finite-sample artifact.

4 Conditional theorem

Conjecture 4.1 (Emergent GUE)**.** *The spacing ratio $\langle \tilde{r} \rangle$ of Hecke eigenvalue angles in $S_k(\mathrm{SL}_2(\mathbb{Z}))$ converges to the GUE value ≈ 0.602 as $k \to \infty$.*

Theorem 4.2 (Conditional convergence to GUE)**.** *Assume the following decorrelation hypothesis: for each fixed k, the normalized Hecke eigenvalues $\{a_p(p)\}_{p \le X}$ of the newforms in S_k satisfy, for any fixed m and distinct primes $p_1, \ldots, p_m$,*

$$\mathbb{E}\Big[\prod_{j=1}^{m} f(\theta_{p_j})\Big] \longrightarrow \prod_{j=1}^{m} \mathbb{E}[f(\theta_{p_j})] \quad \text{as } k \to \infty$$

for suitable test functions f, where the expectation is over the ensemble of newforms. Then the spacing ratio $\langle \tilde{r} \rangle$ converges to the GUE value as $k \to \infty$.

Proof sketch. The decorrelation hypothesis implies that the eigenvalue angles become asymptotically independent as the dimension grows. By the Wigner–Dyson universality theorem, independent eigenvalue angles with the Sato–Tate single-point distribution produce GUE spacing statistics in the large-dimension limit. The key step is verifying that the Sato–Tate distribution on individual angles, combined with asymptotic independence, is sufficient to activate the universality mechanism. □

Remark 4.3. This conditional theorem reduces the "emergent GUE" conjecture to a decorrelation property of Hecke eigenvalues in high-dimensional spaces—a problem in analytic number theory that may be approachable via the theory of automorphic L-functions and the Katz–Sarnak equidistribution framework.

5 Discussion

5.1 Connection to existing results

The Sato–Tate conjecture (proved by Barnet-Lamb, Geraghty, Harris, and Taylor) establishes that individual Hecke eigenvalues of non-CM modular forms follow the Sato–Tate distribution $\frac{2}{\pi}\sin^2\theta$. Our observation that the *spacing ratio* evolves from Sato–Tate (~ 0.33) toward GOE (~ 0.47) as the space dimension grows is a genuinely new finding, not predicted by the existing Sato–Tate theorem.

The Katz–Sarnak theory predicts that zero-spacing statistics of L-function families converge to random matrix universality classes. Our result can be viewed as a finite-dimensional precursor of this prediction: the "family" is the set of Hecke operators $\{T_p\}$ acting on a fixed space S_k, and the "dimension" parameter is k (or equivalently $\dim S_k$).

5.2 Limitations and future work

1. **Precision.** The current computation uses 50-digit mpmath arithmetic, which introduces Ramanujan violations at higher weights. SageMath's modular symbols computation would provide exact rational eigenvalues.
2. **Sample size.** Only 20 primes are used, yielding 40–100 angles. Increasing to 50+ primes would improve statistical power significantly.
3. **Higher dimensions.** Computing $k = 60$ ($d = 7$), $k = 72$ ($d = 8$), and $k = 84$ ($d = 9$) would determine whether the upward trend continues or saturates.
4. **Characteristic polynomial zeros.** The true GUE is expected in the zeros of $\det(xI - T_p)$, not in the eigenvalue angles. Computing these zeros requires a different computational approach.

6 Conclusion

We have reported the first numerical observation of emergent level repulsion in families of Hecke operators. The spacing ratio of normalized eigenvalue angles increases from ~ 0.33 (Sato–Tate) at dimension 1 to ~ 0.47 (Poisson–GOE transition) at dimension 4, consistent with the Katz–Sarnak prediction of emergent universality in high-dimensional arithmetic systems.

This result, combined with the conditional theorem linking the GUE limit to a decorrelation hypothesis, provides the first concrete numerical evidence for the emergence of Wigner–Dyson statistics in deterministic arithmetic systems.

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References

[1] N. M. Katz and P. Sarnak, *Zeroes of zeta functions and symmetry*, Bull. Amer. Math. Soc. **36**(1), 1–26, 1999.

[2] [Anonymous], *On the Spectral Structure of Sinc-Kernel Matrices on the Primes*, 2026.

[3] [Anonymous], *Non-Convolution Arithmetic Matrices on the Primes*, 2026.

[4] [Anonymous], *Spectral Statistics of Deterministic Arithmetic Prime Matrices: A Phase Diagram from GOE to Poisson*, 2026.

[5] [Anonymous], *From Semicircle to Sato–Tate: Moment Convergence, Hecke Angle Statistics, and the Dimension of* $S_{24}(\mathrm{SL}_2(\mathbb{Z}))$, 2026.